ORDINARY DIFFERENTIAL EQUATIONS • UNIT 7.01

Systems of Linear ODEs & Phase Plane Topology

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 D	Q2 D	Q3 B	Q4 A	Q5 B
Q6 A	Q7 A	Q8 D	Q9 C	Q10 B
Q11 C	Q12 A	Q13 D	Q14 D	Q15 B
Q16 A	Q17 B	Q18 A	Q19 B	Q20 D
Q21 C	Q22 B	Q23 A	Q24 C	Q25 A

Systems of Linear ODEs & Phase Plane Topology Practice Exercises

1 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $\pm i$. The origin is a:

A saddle point (unstable)

B unstable spiral

C stable node

D **center (stable, not asymptotically)** ✓

WORKED SOLUTION & VERIFICATION

Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: center (stable, not asymptotically).

2 MCQ

Solve $x' = 2x,\ y' = -y$ with $x(0) = -1,\ y(0) = 3$. Find $x(\ln(2))$.

A $-\frac{1}{2}$

B 4

C 12

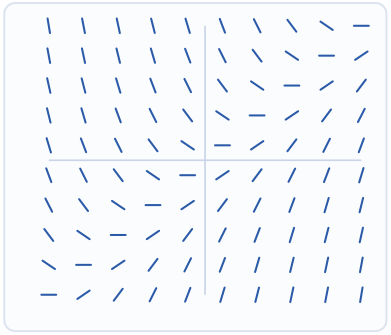
D -4 ✓

WORKED SOLUTION & VERIFICATION

The equations are decoupled: $x = -e^{2t}$, $y = 3e^{-t}$. Hence -4.

3 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + 3y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.



A $\begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$

B $\begin{bmatrix} 0 & 1 \\ -3 & 0 \end{bmatrix}$ ✓

C $\begin{bmatrix} 1 & 0 \\ -3 & 0 \end{bmatrix}$

D $\begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = -3x_1$, so $A = \begin{bmatrix} 0 & 1 \\ -3 & 0 \end{bmatrix}$.

4 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $\pm 5i$. The origin is a:

A **center (stable, not asymptotically)** ✓

B unstable node

C saddle point (unstable)

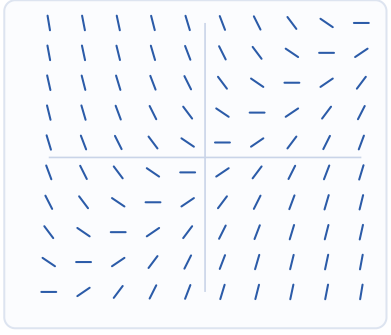
D unstable spiral

WORKED SOLUTION & VERIFICATION

Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: center (stable, not asymptotically).

5 MCQ

Solve $x' = -x,\ y' = -y$ with $x(0) = -2,\ y(0) = 4$. Find $y(\ln(2))$.



A -2

B 2 ✓

C -1

D 4

WORKED SOLUTION & VERIFICATION

The equations are decoupled: $x = -2e^{-t}$, $y = 4e^{-t}$. Hence 2.

6 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + y' + y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- **A** $\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$ ✓
- **B** $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
- **C** $\begin{bmatrix} 1 & 0 \\ -1 & -1 \end{bmatrix}$
- **D** $\begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = -x_1 - x_2$, so $A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$.

7 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $-2 \pm 4i$. The origin is a:

- **A** **stable spiral** ✓
- **B** unstable spiral
- **C** unstable node
- **D** stable node

WORKED SOLUTION & VERIFICATION

Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: stable spiral.

8 MCQ

Solve $x' = -x,\ y' = y$ with $x(0) = 4,\ y(0) = -1$. Find $x(1)$.

- **A** $4e$
- **B** $-\frac{4}{e}$
- **C** $-\frac{1}{e}$
- **D** $\frac{4}{e}$ ✓

WORKED SOLUTION & VERIFICATION

The equations are decoupled: $x = 4e^{-t}$, $y = -e^{t}$. Hence $\frac{4}{e}$.

9 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

- **A** 44
- **B** 43
- **C** **40** ✓
- **D** 24

WORKED SOLUTION & VERIFICATION

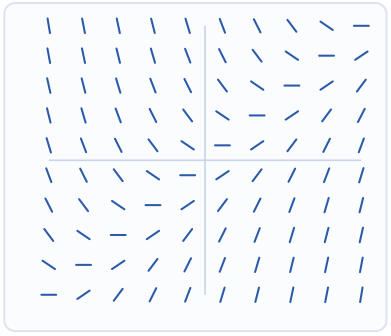
From $\lambda^2 - 40\lambda + 144 = 0$, the sum of roots is 40.

10 MCQ

Consider the system of linear differential equations given by:

$$\begin{cases} \dfrac{dx}{dt} = -x + 2y \\ \dfrac{dy}{dt} = 3x - 2y \end{cases}$$

Let the general solution be of the form $\mathbf{x}(t) = c_1\mathbf{v}_1 e^{\lambda_1 t} + c_2\mathbf{v}_2 e^{\lambda_2 t}$, where $\lambda_1 < \lambda_2$. Determine the value of λ_2.



A 2

B 1 ✓

C 3

D 4

WORKED SOLUTION & VERIFICATION

The given system can be written in matrix form as $\dfrac{d\mathbf{x}}{dt} = A\mathbf{x}$, where $A = \begin{pmatrix} -1 & 2 \\ 3 & -2 \end{pmatrix}$.

To find the eigenvalues λ, we solve the characteristic equation $\det(A - \lambda I) = 0$.

$$\det\begin{pmatrix} -1-\lambda & 2 \\ 3 & -2-\lambda \end{pmatrix} = 0$$

$$(-1-\lambda)(-2-\lambda) - (2)(3) = 0$$

$$(1+\lambda)(2+\lambda) - 6 = 0$$

$$2 + \lambda + 2\lambda + \lambda^2 - 6 = 0$$

$$\lambda^2 + 3\lambda - 4 = 0$$

Factoring the quadratic equation:

$$(\lambda + 4)(\lambda - 1) = 0$$

The eigenvalues are $\lambda_1 = -4$ and $\lambda_2 = 1$.

Since the problem states $\lambda_1 < \lambda_2$, we have $\lambda_1 = -4$ and $\lambda_2 = 1$. The question asks for λ_2.

11 MCQ

The separable initial value problem $\dfrac{du}{ds} = 18s$ satisfies $u(0) = 8$. Determine the solution value $u(7)$.

A 452

B 458

C 449 ✓

D 441

WORKED SOLUTION & VERIFICATION

Integrating gives $u(s) = 9s^2 + 8$, so $u(7) = 449$.

12 MCQ

Consider the system of linear differential equations: $\begin{aligned} \frac{dx}{dt} &= ax + by \\ \frac{dy}{dt} &= cx + dy \end{aligned}$ where a, b, c, d are constants. The characteristic equation for the system is $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$. We are given that the eigenvalues of the system are $\lambda_1 = 2$ and $\lambda_2 = -1$. The trace of the coefficient matrix, $T = a + d$, is 1. Determine the value of the determinant of the coefficient matrix, $D = ad - bc$.

- **A** -2 ✓
- **B** -4
- **C** 2
- **D** 4

WORKED SOLUTION & VERIFICATION

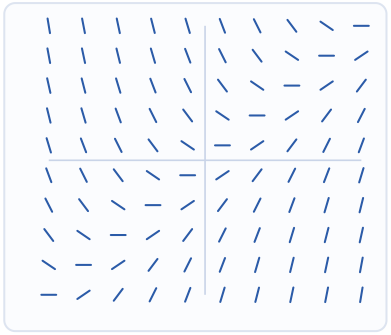
The characteristic equation of a 2x2 system of linear differential equations $\begin{aligned} \frac{dx}{dt} &= ax + by \\ \frac{dy}{dt} &= cx + dy \end{aligned}$ is given by $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$. This can be written in terms of the trace (T = a+d) and determinant (D = ad-bc) of the coefficient matrix as $\lambda^2 - T\lambda + D = 0$. We are given that the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$. The sum of the roots of a quadratic equation $p\lambda^2 + q\lambda + r = 0$ is $-q/p$, and the product of the roots is r/p. For the characteristic equation $\lambda^2 - T\lambda + D = 0$, the sum of the eigenvalues is $\lambda_1 + \lambda_2 = -(-T)/1 = T$, and the product of the eigenvalues is $\lambda_1\lambda_2 = D/1 = D$. We are given that the trace $T = a + d = 1$. Let's verify this with the given eigenvalues: $\lambda_1 + \lambda_2 = 2 + (-1) = 1$, which matches the given trace. We need to find the determinant $D = ad - bc$. Using the product of the eigenvalues: $D = \lambda_1\lambda_2 = (2)(-1) = -2$.

13 MCQ

Consider the system of linear differential equations given by:

$$\frac{dx}{dt} = x + 2y$$
$$\frac{dy}{dt} = 3x + 2y$$

If the solution to this system is of the form $x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ and $y(t) = k_1 c_1 e^{\lambda_1 t} + k_2 c_2 e^{\lambda_2 t}$, what is the value of the larger eigenvalue λ?



A 3

B 6

C 5

D 4 ✓

WORKED SOLUTION & VERIFICATION

The given system can be written in matrix form as $\mathbf{x}'(t) = A\mathbf{x}(t)$, where $A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{x}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$.

To find the eigenvalues λ, we solve the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix.

$\det\begin{pmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{pmatrix} = 0$

$(1-\lambda)(2-\lambda) - (2)(3) = 0$

$2 - \lambda - 2\lambda + \lambda^2 - 6 = 0$

$\lambda^2 - 3\lambda - 4 = 0$

Factoring the quadratic equation:

$(\lambda - 4)(\lambda + 1) = 0$

The eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = -1$. The larger eigenvalue is 4.

14 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + 3y' - 4y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

A $\begin{bmatrix} 1 & 0 \\ 4 & -3 \end{bmatrix}$

B $\begin{bmatrix} 0 & 1 \\ -3 & 4 \end{bmatrix}$

C $\begin{bmatrix} 0 & 1 \\ -4 & 3 \end{bmatrix}$

D $\begin{bmatrix} 0 & 1 \\ 4 & -3 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = 4x_1 - 3x_2$, so $A = \begin{bmatrix} 0 & 1 \\ 4 & -3 \end{bmatrix}$.

15 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues -5 and 3. The origin is a:

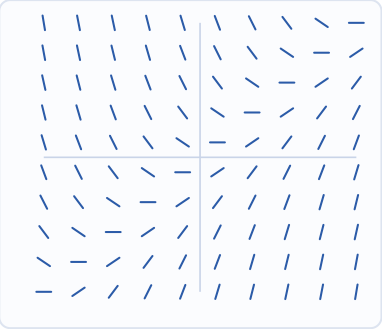
- A stable node
- **B saddle point (unstable)** ✓
- C stable spiral
- D center (stable, not asymptotically)

WORKED SOLUTION & VERIFICATION

Real eigenvalues of opposite sign give a saddle; same sign gives a node (stable if negative). Here: saddle point (unstable).

16 MCQ

Solve $x' = x,\ y' = 2y$ with $x(0) = 3,\ y(0) = -2$. Find $x(2)$.



- **A** $3e^2$ ✓
- B $-2e^2$
- C $3e^4$
- D $-3e^2$

WORKED SOLUTION & VERIFICATION

The equations are decoupled: $x = 3e^t$, $y = -2e^{2t}$. Hence $3e^2$.

17 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' - 5y' + y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- A $\begin{bmatrix} 0 & 1 \\ 1 & -5 \end{bmatrix}$
- **B** $\begin{bmatrix} 0 & 1 \\ -1 & 5 \end{bmatrix}$ ✓
- C $\begin{bmatrix} 1 & 0 \\ -1 & 5 \end{bmatrix}$
- D $\begin{bmatrix} 0 & 1 \\ 5 & -1 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = -x_1 + 5x_2$, so $A = \begin{bmatrix} 0 & 1 \\ -1 & 5 \end{bmatrix}$.

18 MCQ

In this problem, the matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues -3 and 2. The origin is a:

- **A** saddle point (unstable) ✓
- **B** stable spiral
- **C** center (stable, not asymptotically)
- **D** unstable node

WORKED SOLUTION & VERIFICATION

Real eigenvalues of opposite sign give a saddle; same sign gives a node (stable if negative). Here: saddle point (unstable).

19 MCQ

Solve $x' = 3x,\ y' = y$ with $x(0) = -1,\ y(0) = 4$. Find $x(1)$.

- **A** $-e$
- **B** $-e^3$ ✓
- **C** e^3
- **D** $4e^3$

WORKED SOLUTION & VERIFICATION

The equations are decoupled: $x = -e^{3t}$, $y = 4e^t$. Hence $-e^3$.

20 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + 2y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- **A** $\begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$
- **B** $\begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$
- **C** $\begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$
- **D** $\begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = -2x_1$, so $A = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}$.

21 MCQ

In this problem, the matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues $\pm 4i$. The origin is a:

- **A** stable spiral
- **B** unstable node
- **C** center (stable, not asymptotically) ✓
- **D** stable node

WORKED SOLUTION & VERIFICATION

Complex eigenvalues give spirals; the sign of the real part decides stability, and a zero real part gives a center. Here: center (stable, not asymptotically).

22 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' - 3y' - 2y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- A $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$
- B $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$ ✓
- C $\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$
- D $\begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = 2x_1 + 3x_2$, so $A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$.

23 MCQ

The matrix of $\mathbf{x}' = A\mathbf{x}$ has eigenvalues 3 and -4. The origin is a:

- A **saddle point (unstable)** ✓
- B unstable spiral
- C stable spiral
- D center (stable, not asymptotically)

WORKED SOLUTION & VERIFICATION

Real eigenvalues of opposite sign give a saddle; same sign gives a node (stable if negative). Here: saddle point (unstable).

24 MCQ

Solve $x' = -2x,\ y' = -3y$ with $x(0) = -2,\ y(0) = -2$. Find $y(2)$.

- A $-\frac{2}{e^4}$
- B $\frac{2}{e^6}$
- C $-\frac{2}{e^6}$ ✓
- D $-\frac{4}{e^6}$

WORKED SOLUTION & VERIFICATION

The equations are decoupled: $x = -2e^{-2t}$, $y = -2e^{-3t}$. Hence $-\frac{2}{e^6}$.

25 MCQ

With $x_1 = y$, $x_2 = y'$, the equation $y'' + 4y' + 3y = 0$ becomes $\mathbf{x}' = A\mathbf{x}$. Find A.

- A $\begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}$ ✓
- B $\begin{bmatrix} 0 & 1 \\ 3 & 4 \end{bmatrix}$
- C $\begin{bmatrix} 1 & 0 \\ -3 & -4 \end{bmatrix}$
- D $\begin{bmatrix} 0 & 1 \\ -4 & -3 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

$x_1' = x_2$ and $x_2' = y'' = -3x_1 - 4x_2$, so $A = \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}$.